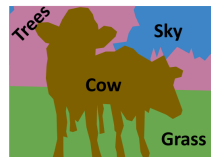
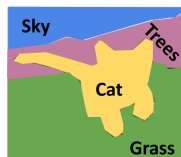


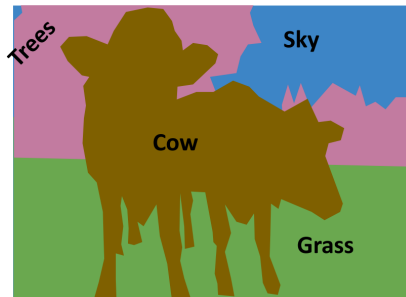
- **Things:** Object categories that can be separated into object instances (e.g. cats, cars, person)
- **Stuff:** Object categories that cannot be separated into instances (e.g. sky, grass, water, trees)



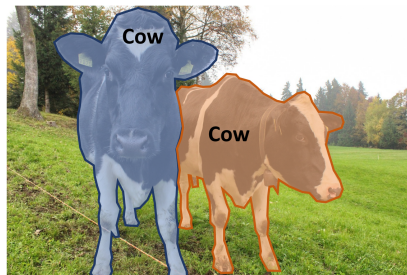
- **Object Detection:** Detects individual object instances, but only gives box(Only things!)



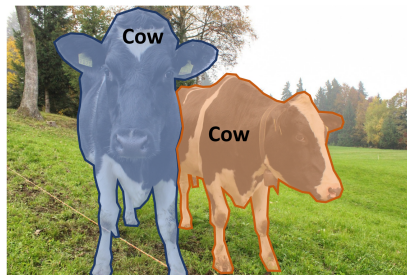
- **Semantic Segmentation:**
Gives per-pixel labels, but merges instances (Both things and stuff)

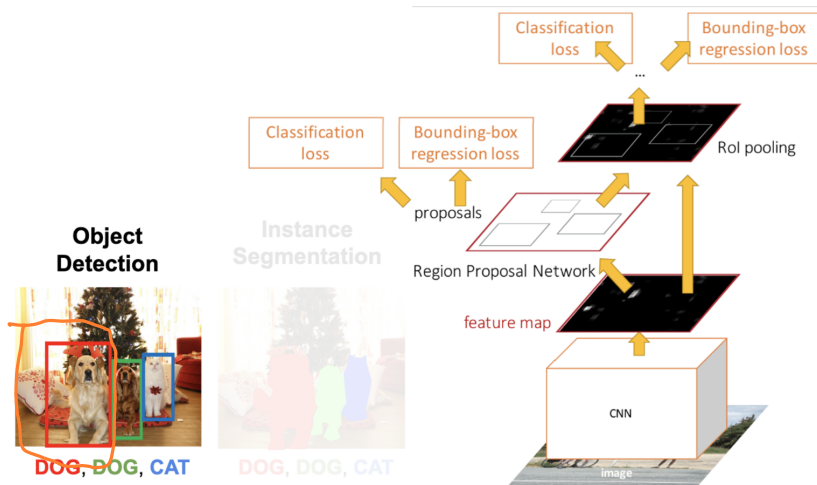


- Detect all objects in the image, and identify the pixels that belong to each object (Only things!)

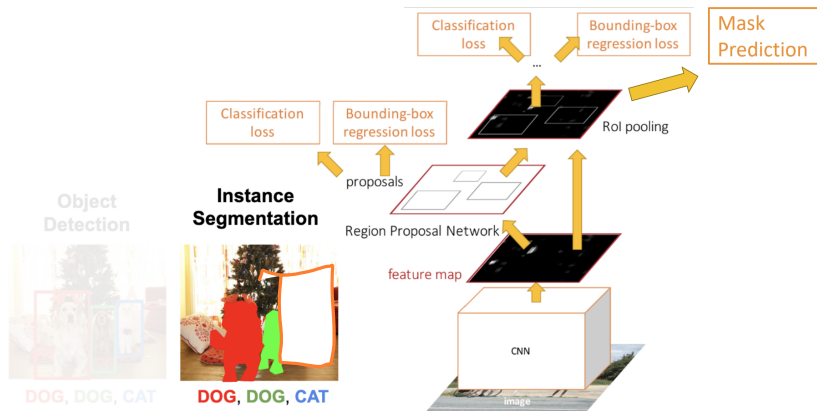


- ▶ Detect all objects in the image, and identify the pixels that belong to each object (Only things!)
- ▶ **Approach:** Perform object detection, then predict a segmentation mask for each object!



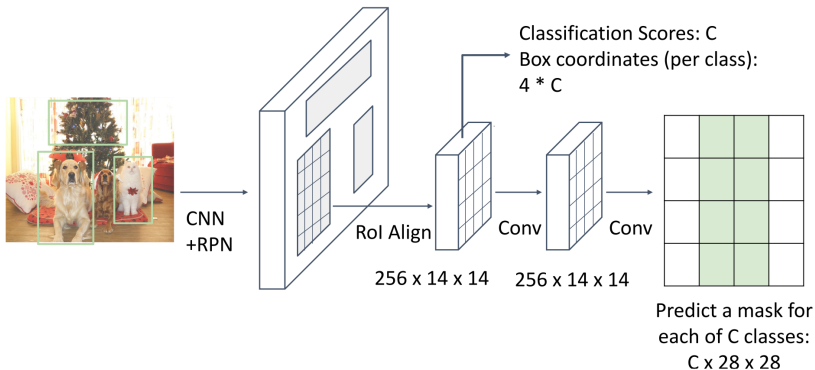


Instance Segmentation: Mask R-CNN

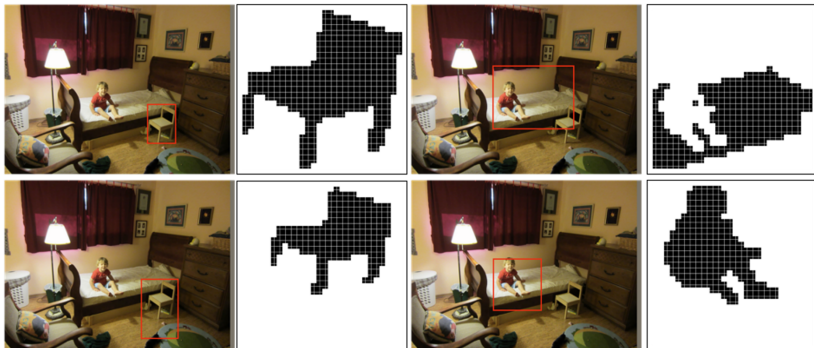


⁰He et al, "Mask R-CNN", ICCV 2017

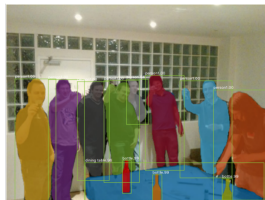
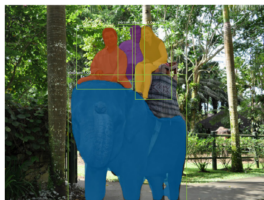
Instance Segmentation: Mask R-CNN



Mask R-CNN: Example Training Targets

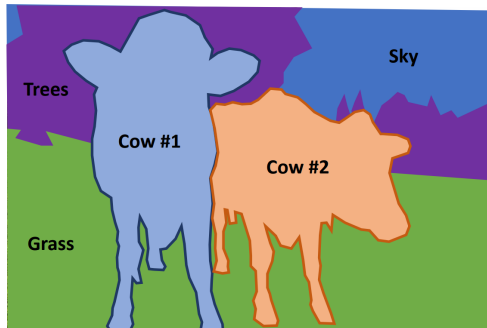


Mask R-CNN: Very Good Results!

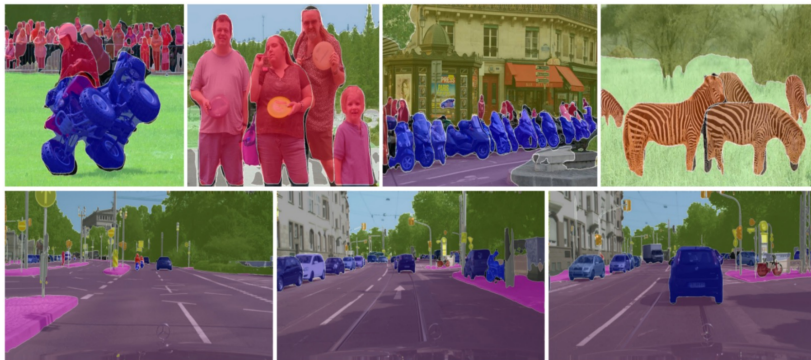


Beyond Instance Segmentation: Panoptic Segmentation

- ▶ Label all pixels in the image (both things and stuff)
- ▶ For "thing" categories also separate into instances

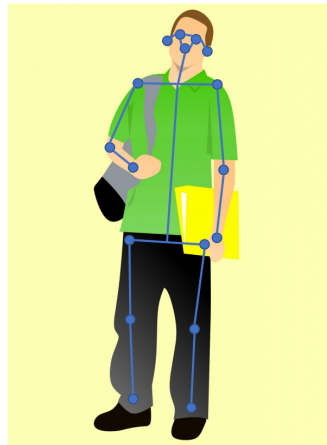


Beyond Instance Segmentation: Panoptic Segmentation

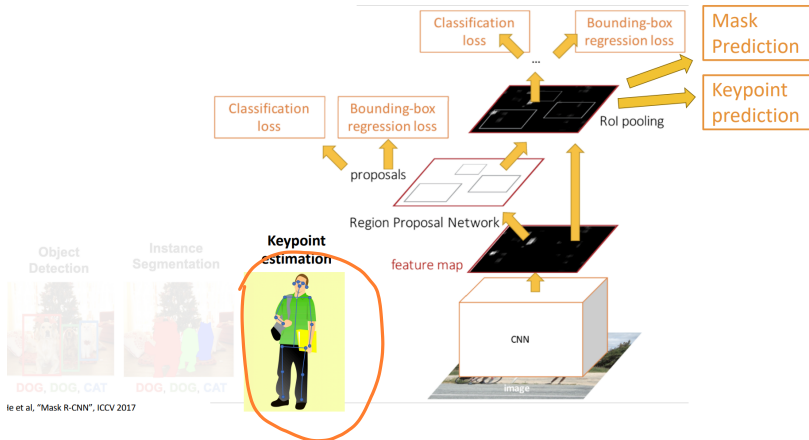


⁰Kirillov et al, "Panoptic Feature Pyramid Networks", CVPR 2019

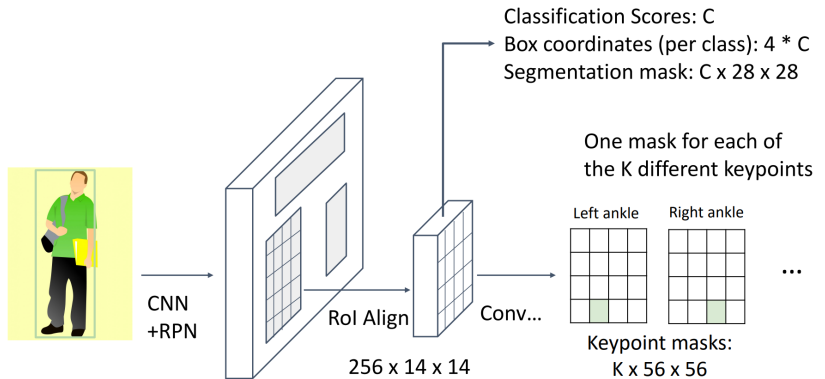
- ▶ Represent the pose of a human by locating a set of keypoint se.g. 17 keypoints:
- ▶ Nose
- ▶ Left / Right eye
- ▶ Left / Right earLeft / Right shoulder
- ▶ Left / Right elbow
- ▶ Left / Right wrist



Mask R-CNN: Keypoint Estimation



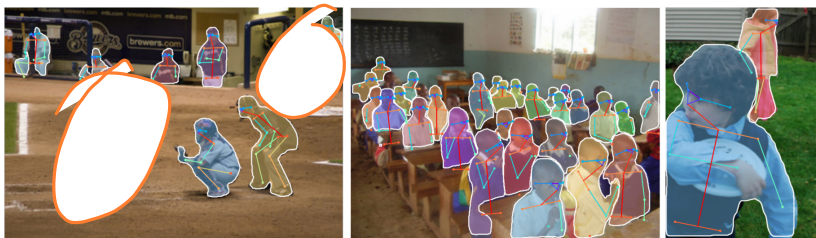
Mask R-CNN: Keypoint Estimation



Do et al. "Mask R-CNN" ICCV 2017

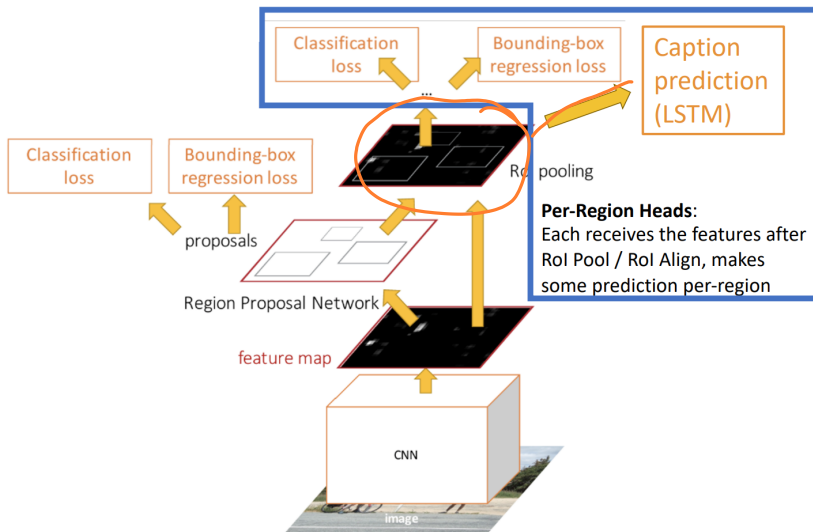
Ground-truth has one "pixel" turned on per keypoint. Train with softmax loss

Joint Instance Segmentation and Pose Estimation

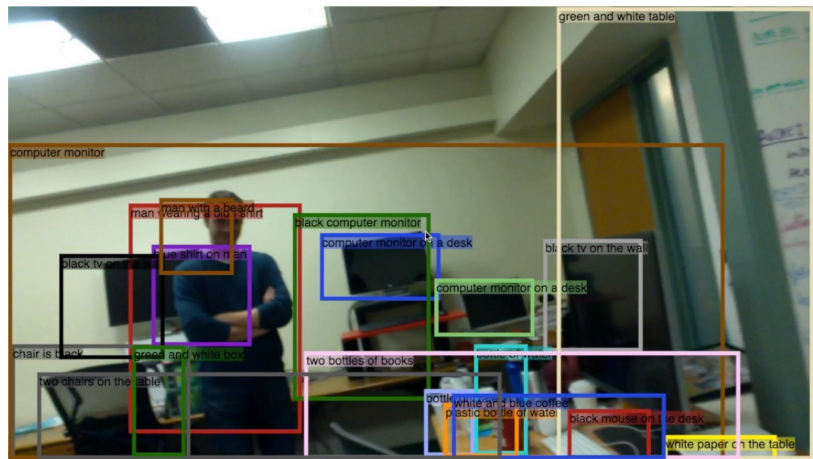


⁰He et al, "Mask R-CNN", ICCV 2017

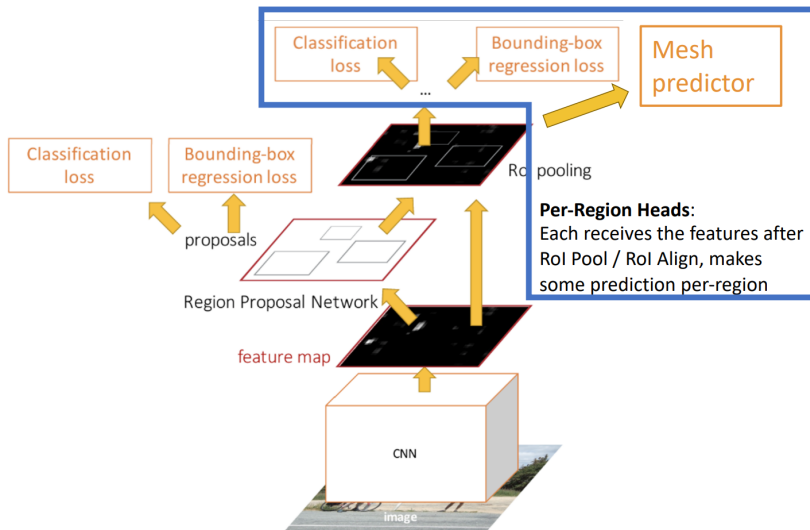
Captioning: Predict a caption per region!



Captioning: Predict a caption per region!

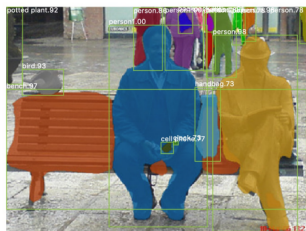


Johnson, Karpathy, and Fei-Fei, "DenseCap: Fully Convolutional Localization Networks for Dense Captioning", CVPR 2016



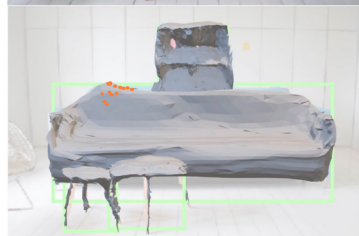
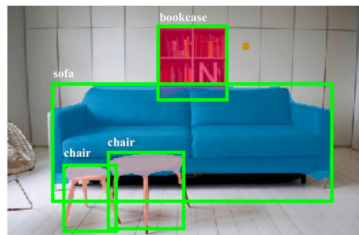
Mask R-CNN:

2D Image -> 2D shapes



Mesh R-CNN:

2D Image -> **3D** shapes



Gkioxari, Malik, and Johnson, "Mesh R-CNN", ICCV 2019



- ▶ **Goal:** Track objects over a sequence of photos or a video
- ▶ Exceedingly challenging in multi-object tracking scenarios
- ▶ Need to take care of not mixing up or losing objects midway
- ▶ **One Solution:** Perform object detection and assign IDs to each object and store its feature vector. Then track the objects based on its ID and feature vector

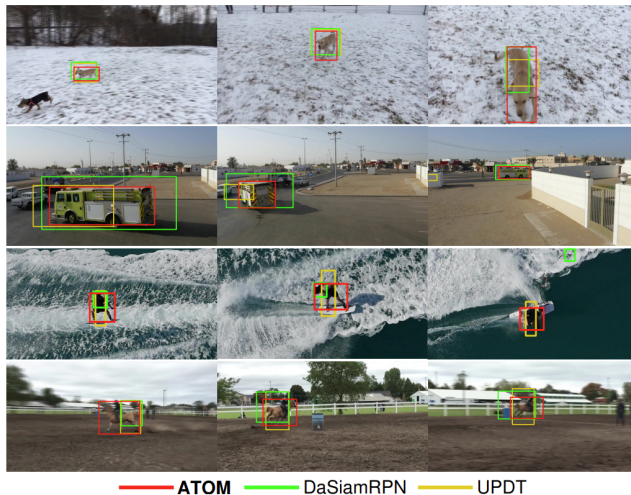


Figure 2: Comparison of 3 approaches for object tracking

Danelljan et al.

These slides have been adapted from

- ▶ Fei-Fei Li, Yunzhu Li & Ruohan Gao, Stanford CS231n: **Deep Learning for Computer Vision**
- ▶ Assaf Shocher, Shai Bagon, Meirav Galun & Tali Dekel, WAIC DL4CV **Deep Learning for Computer Vision: Fundamentals and Applications**
- ▶ Justin Johnson, UMich EECS 498.008/598.008: **Deep Learning for Computer Vision**